Activity 3 - Diagnose and Prioritise the Pilot's Quality Failure

Course: Effective Project Management for Small Projects (TGS-2020505545) Topic: 3 | Apply Methods and Tools to Drive a Project Learning outcome: LO3 Duration: 75 minutes Team size: 3-4 learners Tool set: Fishbone, Check Sheet, 5 Whys, Pareto and Control Chart

---

Objective

Use appropriate methods and tools to drive progress and improve quality (A3, K3).

The situation

During outlet testing, only 82% of orders are ready at the promised collection time and pick accuracy has fallen to 94%. The software vendor blames store staff; store staff blame late inventory updates. The sponsor wants action before the team has separated symptoms, causes and normal variation.

What you will do

Capture defect facts, map potential causes, drill to an actionable root cause, rank the vital few, interpret process behaviour and commit one measurable improvement action.

What you will produce

A reconciled check sheet, fishbone diagram, five-level causal chain, Pareto chart, control-chart interpretation and a SMART corrective experiment with owner and review date.

Tool files in this folder

- `tools/35 Check Sheets.xlsx`
- `tools/37 Ishikawa Diagram.xlsx`
- `tools/38 Five Whys.xlsx`
- `tools/36 Pareto Principle Template.xlsx`
- `tools/34 Control Chart.xlsx`

Data files

- `data/quality-test-data.csv`

Step-by-step instructions

1. Open 35 Check Sheets.xlsx and enter the supplied 60-order test dataset by defect type, outlet, shift and date.
2. Reconcile the total. Separate late-ready orders from wrong-item, stock-out, notification and payment defects.
3. Open the Fishbone browser tool or 37 Ishikawa Diagram.xlsx. Set the effect to 'Orders not ready accurately at the promised time'.
4. Generate causes across People, Process, Technology, Data, Measurement and Environment. Mark facts with F and hypotheses with H.

5. Select the highest-frequency cause family and complete 38 Five Whys.xlsx until the root is a process or system cause the team can influence.

6. Read the chain backwards with 'because'. Repair any causal jump or circular statement.

7. Open 36 Pareto Principle Template.xlsx and rank the defect categories. Add the cumulative percentage and identify the crossing of 80%.

8. Use 34 Control Chart.xlsx to plot daily ready-on-time percentage. Decide whether the signal suggests a stable process or special-cause behaviour.

9. Propose one SMART improvement experiment with an owner, target, completion date and measure. Allocate it on the project Kanban board.

10. Explain to the sponsor why fixing every listed cause at once would be slower and less controlled than testing the vital few.

Evidence to submit

Complete WORKSHEET.md and attach/export the completed tool files named above.

Acceptance criteria

The data total reconciles; Pareto reaches 100%; the five-whys chain ends at an actionable cause; the control chart interpretation matches the pattern; and the corrective experiment has an owner, measure and review date.

Debrief

- Fishbone broadens the search, 5 Whys deepens one chain, and Pareto tells you which chain is worth funding first.
- A control limit describes process behaviour; it is not the same as a customer specification limit.
- Improvement becomes project work only when it has capacity, ownership and a verification date.

The full procedure is also reproduced in the Learner Guide, Activity 3. The trainer deck contains only the briefing and success criteria.